Yizhou Ding (Jason)

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Education

Université Paris Sciences & Lettres Sep 2021 – Jul 2024
Artificial Intelligence | *Master's Degree (Second Degree)*

Huazhong University of Science and Technology Sep 2021 – Jul 2024
New Energy Science and Engineering | *Master's Degree* | National Scholarship (Top 1%)

Wuhan University of Technology (Project 211) Sep 2017 – Jul 2021
Communication Engineering (Excellence Program) | *Bachelor's Degree*

Professional Experience

Alibaba Group – Zhejiang Tmall Technology Co., Ltd. Jul 2024 – Aug 2026
AI Algorithm Engineer | Performance Ratings: E & A+

Shenzhen InnoX Academy (Startup Incubator) Jan 2024 – Jun 2024
Junior Product Manager Intern

Project Experience

Sales Forecasting with Qwen3 Structured Reasoning and Post-Training
Alibaba | Marketing Algorithms Team
Combined time-series modeling with LLM-based business reasoning to improve sales forecasting under promotional campaigns and price changes, supporting promotional pricing.

- **Time-Series Representation Learning and Model Design:** Led research on Transformer rank collapse, using attention stable rank and representation cosine similarity to diagnose degradation in shallow layers and representation homogenization under strongly autocorrelated inputs. Proposed DEFormer, combining raw-series and first-difference encoding, a `log1p` target transformation, multitask regression and classification, and SubAttention; validated the components through ablation studies. Reduced weighted mean absolute percentage error (wMAPE) by 12.7% and 5.6% relative to PatchTST, the strongest baseline, on two real-world retail datasets; documented the work in a research manuscript.
- **Task-Aligned SFT and Representation Fusion:** Addressed the limitations of static semantic representations in capturing sales responses to changing marketing conditions. Constructed six supervision classes from observed sales-change directions and magnitudes, fine-tuned Qwen3-4B with LoRA, and fused the task-adapted representations into the downstream forecasting network. Reduced offline wMAPE by 3.7% overall and 6.2% on major-promotion samples with sharp sales increases, relative to the frozen-representation baseline.
- **Reinforcement Learning with Verifiable Rewards (RLVR):** Applied GRPO to the SFT model to expand reasoning-path exploration, using the correctness of the final sales-change label as a verifiable reward. Refined business judgments through sampled reasoning paths and outcome feedback, then integrated the resulting representations into the price–sales model. Reduced offline wMAPE by 2.9% relative to the SFT-representation baseline.
- **On-Policy Distillation (OPD):** Addressed sparse outcome-level rewards, limited reasoning-process supervision, and persistent misclassifications in RLVR. Used a Qwen3-27B teacher to provide token-level probability-distribution supervision on student-sampled trajectories, improving chain-of-thought reasoning and the consistency of sales-trend predictions. Retained the same representation-fusion and downstream-retraining pipeline, achieving a further 1.1% relative reduction in offline wMAPE.
- **Project-Level Online A/B Results:** Across successive project iterations, achieved cumulative gains of approximately 5.06% in gross merchandise value (GMV), 5.50% in sub-order count, and 4.51% in GMV coverage relative to the initial version, supporting promotional pricing and subsidy allocation.

Operations Agent for High-Potential Product Growth and Tier Retention
Alibaba | Marketing Algorithms Team
Combined price–sales forecasting with agent-based planning to develop high-potential products across campaign periods and maintain top-selling products' tiers in the primary search channel, supporting operational efficiency and long-term GMV growth.

- **Tool Integration and Product Assessment:** Exposed existing price–sales forecasting models as callable tools. Integrated promotion, advertising, and search-exposure signals to identify high-potential products and tier-downgrade risks, assess growth potential and retention needs, and support differentiated operating strategies.
- **LangGraph Orchestration and Persistent Memory:** Orchestrated batch operations workflows with LangGraph and used Qwen3.5-35B-A3B to generate plans spanning multiple campaign periods. Built SKU-level persistent memory in SQL, with separate tables for current memory summaries and operational histories. Retrieved feedback by product, scenario, and round before planning, injected it into context, and updated memory after feedback collection to inform subsequent plans.
- **Joint Price and Budget Decision Design:** Designed a joint modeling framework based on GMV(Budget, Price), aligning price–sales and advertising-budget–GMV models to support promotional pricing and advertising-budget allocation under shared-budget and ROI constraints.
- **Reliability and Human Review:** Implemented retry and fallback mechanisms for tool-call and workflow failures. Incorporated human review of operational recommendations, combining automated tool execution with human oversight for plan assessment and exception handling.
- **Two-Phase Online A/B Evaluation: Phase 1:** Compared model-assisted operations against manual operations, distributing the overall operating target evenly across campaign periods. Recommendation adoption reached 26.6%, and treatment-group GMV growth was 65.2% higher than control-group growth. **Phase 2:** Used the Phase 1 model-assisted approach as the control and introduced an agent workflow to generate plans around period-specific targets. Adoption reached 39.3%, and treatment-group GMV growth was 27.7% higher than control-group growth.

Smart-Ring Product Development
Shenzhen InnoX Academy | Startup Team | Product Manager

- **Product Definition and Prototyping:** Led development from concept to first-generation prototype. Translated market research and 200+ user interviews into prioritized health-monitoring requirements, core feature plans, prototypes, and technical feasibility assessments, working with software and hardware engineers.
- **Iteration and Cross-Functional Delivery:** Built a community of 200+ early adopters and organized prototype testing, feedback collection, and agile iterations. Coordinated internal engineering and external operations resources to deliver core milestones within six months; the project secured seed funding from the incubator.

Selected Publications

Selected work appeared in *Nature Reviews Electrical Engineering* and IEEE Transactions journals. The complete list is available on Google Scholar.

1. "DEFormer: Understanding and Alleviating Rank Collapse Caused by Autocorrelation in Transformer-based Time Series Forecaster." *Manuscript submitted*, 2026.
2. "Privacy-Preserving Graph Inference Network for Multi-Entity Wind Power Forecast: A Federated Learning Approach." *IEEE Transactions on Network Science and Engineering*, 2025.
3. "Artificial Intelligence-Based Methods for Renewable Power System Operation." *Nature Reviews Electrical Engineering*, 2024.
4. "Dense Skip Attention Based Deep Learning for Day-Ahead Electricity Price Forecasting." *IEEE Transactions on Power Systems*, 2023.

Technical Skills

Programming and ML: Python, SQL, PyTorch, TensorFlow, Hugging Face, Time-Series Modeling
Large Language Models: Qwen3, Transformers, LoRA SFT, RLVR (GRPO), On-Policy Distillation (OPD)
Languages: English (IELTS 7.5/9)

Honors and Awards

National Competition Awards (4): Two First Prizes (highest placement: Top 0.9%) and two Third Prizes 2019 – 2023
National Scholarship: One award 2024